Equations

CHEMISTRY REFERENCE

Measurement

Density	$D = \frac{m}{V}$
Percent error	$\%\ \text{error} = \frac{\lvert \text{experimental} - \text{accepted} \rvert}{\text{accepted}} \times 100$
Percent composition	$\%\ \text{element} = \frac{\text{mass of element}}{\text{mass of compound}} \times 100$
Average atomic mass	$(\text{mass}_1 \times \text{abundance}_1) + (\text{mass}_2 \times \text{abundance}_2) + \ldots$ abundance as a decimal

The Mole

Moles from mass	$n = \frac{m}{M}$ n moles · m mass (g) · M molar mass (g/mol)
Particles	$\text{particles} = n \times N_A$

Gas Laws T in kelvin

Boyle's law	$P_1V_1 = P_2V_2$
Charles's law	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
Gay-Lussac's law	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
Combined gas law	$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$
Avogadro's law	$\frac{V_1}{n_1} = \frac{V_2}{n_2}$
Ideal gas law	$PV = nRT$
Dalton's law	$P_{\text{total}} = P_1 + P_2 + P_3 + \ldots$

Solutions

Molarity	$M = \frac{\text{moles of solute}}{\text{liters of solution}}$
Dilution	$M_1V_1 = M_2V_2$

Thermochemistry

Heat	$q = mc\Delta T$
Temperature change	$\Delta T = T_{\text{final}} - T_{\text{initial}}$

Acids & Bases

pH	$\text{pH} = -\log[H^+] \quad [H^+] = 10^{-\text{pH}}$
pOH	$\text{pOH} = -\log[OH^-] \quad [OH^-] = 10^{-\text{pOH}}$
pH and pOH	$\text{pH} + \text{pOH} = 14$ at 25 °C
Titration (1 : 1)	$M_aV_a = M_bV_b$

Light

Wave speed	$c = \lambda\nu$
Photon energy	$E = h\nu$

Nuclear

Half-life	$\text{amount left} = \text{initial amount} \times (\frac{1}{2})^n$
	$n = \frac{\text{time elapsed}}{\text{half-life}}$

Constants & Conversions

CHEMISTRY REFERENCE

Constants

Avogadro's number	$N_A = 6.022 \times 10^{23}$ particles/mol
Gas constant	R = 0.0821 L·atm/(mol·K)
	R = 8.314 L·kPa/(mol·K)
	R = 62.4 L·mmHg/(mol·K)
Speed of light	$c = 3.00 \times 10^{8}$ m/s
Planck's constant	$h = 6.626 \times 10^{-34}$ J·s
Specific heat of water	c = 4.184 J/(g·°C)
STP	0 °C (273 K) and 1 atm
Molar volume at STP	22.4 L/mol

Metric Prefixes

Prefix	Symbol	Factor
giga	**G**	10^{9}
mega	**M**	10^{6}
kilo	**k**	10^{3}
— base unit —		1
deci	**d**	10^{-1}
centi	**c**	10^{-2}
milli	**m**	10^{-3}
micro	**μ**	10^{-6}
nano	**n**	10^{-9}

Conversions

Pressure	1 atm = 760 mmHg = 760 torr = 101.3 kPa
Temperature	K = °C + 273
Volume	1 mL = 1 cm^3 1 L = 1000 mL
Length	1 in = 2.54 cm
Mass	1 lb = 453.6 g 1 kg = 1000 g
Energy	1 cal = 4.184 J 1 kJ = 1000 J

Dimensional Analysis

$$\boxed{\text{given}\\ \text{number + unit}} \times \frac{\boxed{\text{new unit}}}{\boxed{\text{unit to cancel}}} \times \frac{\boxed{\text{new unit}}}{\boxed{\text{unit to cancel}}} = \boxed{\text{answer}\\ \text{number + unit}}$$

1. Write what you are given, with its unit.
2. Multiply by a conversion factor written so the unit you have is on the bottom.
3. Cancel units. Repeat until only the unit you want is left.
4. Multiply across the top, divide by everything on the bottom.

Example $1.75 \text{ mol } H_2O \times \frac{18.02 \text{ g } H_2O}{1 \text{ mol } H_2O} = 31.5 \text{ g } H_2O$

Significant Figure Rules

CHEMISTRY REFERENCE

Which digits count

Rule	Example	Sig figs
Nonzero digits are always significant.	4.72	**3**
Zeros between nonzero digits are significant.	2005	**4**
Leading zeros are never significant. They only place the decimal point.	0.0032	**2**
Trailing zeros are significant when the number has a decimal point.	3.200 · 150.	**4 · 3**
Trailing zeros with no decimal point are not significant.	1500	**2**
In scientific notation, every digit in the coefficient is significant.	4.50×10^3	**3**
Exact numbers (counted items, defined conversions) have unlimited significant figures.	12 eggs · 1 m = 100 cm	**∞**

Calculations

Multiplying and dividing

The answer has the same number of **significant figures** as the measurement with the fewest.

$2.5\ \text{cm} \times 3.47\ \text{cm} = 8.675 \rightarrow \mathbf{8.7\ cm^2}$

2.5 has 2 significant figures, so the answer has 2.

Adding and subtracting

The answer has the same number of **decimal places** as the measurement with the fewest.

$12.11\ \text{g} + 3.2\ \text{g} = 15.31 \rightarrow \mathbf{15.3\ g}$

3.2 has 1 decimal place, so the answer has 1.

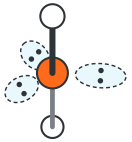
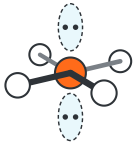
Rounding

Look at the first digit you are dropping. 5 or more: round up. Less than 5: leave the last kept digit alone. 3.476 to 3 sig figs → 3.48 · 3.474 → 3.47

In a multi-step problem, keep extra digits until the final answer, then round once.

VSEPR Shapes

CHEMISTRY REFERENCE

	0 lone pairs	1 lone pair	2 lone pairs	3 lone pairs
2 ELECTRON DOMAINS ELECTRON GEOMETRY **Linear**	 **Linear** 2 bonding · 0 lone | 180° | AX_2			
3 ELECTRON DOMAINS ELECTRON GEOMETRY **Trigonal Planar**	 **Trigonal Planar** 3 bonding · 0 lone | 120° | AX_3	 **Bent** 2 bonding · 1 lone | <120° | AX_2E_1		
4 ELECTRON DOMAINS ELECTRON GEOMETRY **Tetrahedral**	 **Tetrahedral** 4 bonding · 0 lone | 109.5° | AX_4	 **Trigonal Pyramidal** 3 bonding · 1 lone | ≈107° | AX_3E_1	 **Bent** 2 bonding · 2 lone | ≈104.5° | AX_2E_2	
5 ELECTRON DOMAINS ELECTRON GEOMETRY **Trigonal Bipyramidal**	 **Trigonal Bipyramidal** 5 bonding · 0 lone | 90°, 120° | AX_5	 **Seesaw** 4 bonding · 1 lone | <90°, <120° | AX_4E_1	 **T-Shaped** 3 bonding · 2 lone | <90° | AX_3E_2	 **Linear** 2 bonding · 3 lone | 180° | AX_2E_3
6 ELECTRON DOMAINS ELECTRON GEOMETRY **Octahedral**	 **Octahedral** 6 bonding · 0 lone | 90° | AX_6	 **Square Pyramidal** 5 bonding · 1 lone | <90° | AX_5E_1	 **Square Planar** 4 bonding · 2 lone | 90° | AX_4E_2	

central atom (A) ○ bonded atom (X) ◌ lone pair (E) grey bond = pointing behind the page

Electron domains = bonded atoms + lone pairs on the central atom. A double or triple bond counts as one domain.

Periodic Table of the Elements

CHEMISTRY REFERENCE

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 **H** Hydrogen 1.01																	2 **He** Helium 4.00
2	3 **Li** Lithium 6.94	4 **Be** Beryllium 9.01											5 **B** Boron 10.81	6 **C** Carbon 12.01	7 **N** Nitrogen 14.01	8 **O** Oxygen 16.00	9 **F** Fluorine 19.00	10 **Ne** Neon 20.18
3	11 **Na** Sodium 22.99	12 **Mg** Magnesium 24.31											13 **Al** Aluminum 26.98	14 **Si** Silicon 28.09	15 **P** Phosphorus 30.97	16 **S** Sulfur 32.06	17 **Cl** Chlorine 35.45	18 **Ar** Argon 39.95
4	19 **K** Potassium 39.10	20 **Ca** Calcium 40.08	21 **Sc** Scandium 44.96	22 **Ti** Titanium 47.87	23 **V** Vanadium 50.94	24 **Cr** Chromium 52.00	25 **Mn** Manganese 54.94	26 **Fe** Iron 55.85	27 **Co** Cobalt 58.93	28 **Ni** Nickel 58.69	29 **Cu** Copper 63.55	30 **Zn** Zinc 65.38	31 **Ga** Gallium 69.72	32 **Ge** Germanium 72.63	33 **As** Arsenic 74.92	34 **Se** Selenium 78.97	35 **Br** Bromine 79.90	36 **Kr** Krypton 83.80
5	37 **Rb** Rubidium 85.47	38 **Sr** Strontium 87.62	39 **Y** Yttrium 88.91	40 **Zr** Zirconium 91.22	41 **Nb** Niobium 92.91	42 **Mo** Molybdenum 95.95	43 **Tc** Technetium [97]	44 **Ru** Ruthenium 101.07	45 **Rh** Rhodium 102.91	46 **Pd** Palladium 106.42	47 **Ag** Silver 107.87	48 **Cd** Cadmium 112.41	49 **In** Indium 114.82	50 **Sn** Tin 118.71	51 **Sb** Antimony 121.76	52 **Te** Tellurium 127.60	53 **I** Iodine 126.90	54 **Xe** Xenon 131.29
6	55 **Cs** Cesium 132.91	56 **Ba** Barium 137.33	57* **La** Lanthanum 138.91	72 **Hf** Hafnium 178.49	73 **Ta** Tantalum 180.95	74 **W** Tungsten 183.84	75 **Re** Rhenium 186.21	76 **Os** Osmium 190.23	77 **Ir** Iridium 192.22	78 **Pt** Platinum 195.08	79 **Au** Gold 196.97	80 **Hg** Mercury 200.59	81 **Tl** Thallium 204.38	82 **Pb** Lead 207.20	83 **Bi** Bismuth 208.98	84 **Po** Polonium [209]	85 **At** Astatine [210]	86 **Rn** Radon [222]
7	87 **Fr** Francium [223]	88 **Ra** Radium [226]	89** **Ac** Actinium [227]	104 **Rf** Rutherfordiur [267]	105 **Db** Dubnium [268]	106 **Sg** Seaborgium [269]	107 **Bh** Bohrium [270]	108 **Hs** Hassium [271]	109 **Mt** Meitnerium [278]	110 **Ds** Darmstadtiun [281]	111 **Rg** Roentgenium [282]	112 **Cn** Copernicium [285]	113 **Nh** Nihonium [286]	114 **Fl** Flerovium [289]	115 **Mc** Moscovium [290]	116 **Lv** Livermorium [293]	117 **Ts** Tennessine [294]	118 **Og** Oganesson [294]

* Lanthanides	58 **Ce** Cerium 140.12	59 **Pr** Praseodymiui 140.91	60 **Nd** Neodymium 144.24	61 **Pm** Promethium [145]	62 **Sm** Samarium 150.36	63 **Eu** Europium 151.96	64 **Gd** Gadolinium 157.25	65 **Tb** Terbium 158.93	66 **Dy** Dysprosium 162.50	67 **Ho** Holmium 164.93	68 **Er** Erbium 167.26	69 **Tm** Thulium 168.93	70 **Yb** Ytterbium 173.05	71 **Lu** Lutetium 174.97
** Actinides	90 **Th** Thorium 232.04	91 **Pa** Protactinium 231.04	92 **U** Uranium 238.03	93 **Np** Neptunium [237]	94 **Pu** Plutonium [244]	95 **Am** Americium [243]	96 **Cm** Curium [247]	97 **Bk** Berkelium [247]	98 **Cf** Californium [251]	99 **Es** Einsteinium [252]	100 **Fm** Fermium [257]	101 **Md** Mendelevium [258]	102 **No** Nobelium [259]	103 **Lr** Lawrencium [266]

atomic number
Symbol
Name
average atomic mass

A mass in brackets [ ] is the mass number of the element's longest-lived isotope. These elements have no standard atomic mass.

Electronegativity

CHEMISTRY REFERENCE

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 2.20																	2 He
2	3 Li 0.98	4 Be 1.57											5 B 2.04	6 C 2.55	7 N 3.04	8 O 3.44	9 F 3.98	10 Ne
3	11 Na 0.93	12 Mg 1.31											13 Al 1.61	14 Si 1.90	15 P 2.19	16 S 2.58	17 Cl 3.16	18 Ar
4	19 K 0.82	20 Ca 1.00	21 Sc 1.36	22 Ti 1.54	23 V 1.63	24 Cr 1.66	25 Mn 1.55	26 Fe 1.83	27 Co 1.88	28 Ni 1.91	29 Cu 1.90	30 Zn 1.65	31 Ga 1.81	32 Ge 2.01	33 As 2.18	34 Se 2.55	35 Br 2.96	36 Kr
5	37 Rb 0.82	38 Sr 0.95	39 Y 1.22	40 Zr 1.33	41 Nb 1.60	42 Mo 2.16	43 Tc 1.90	44 Ru 2.20	45 Rh 2.28	46 Pd 2.20	47 Ag 1.93	48 Cd 1.69	49 In 1.78	50 Sn 1.96	51 Sb 2.05	52 Te 2.10	53 I 2.66	54 Xe
6	55 Cs 0.79	56 Ba 0.89	57* La 1.10	72 Hf 1.30	73 Ta 1.50	74 W 2.36	75 Re 1.90	76 Os 2.20	77 Ir 2.20	78 Pt 2.28	79 Au 2.54	80 Hg 2.00	81 Tl 1.62	82 Pb 2.33	83 Bi 2.02	84 Po 2.00	85 At 2.20	86 Rn
7	87 Fr 0.70	88 Ra 0.90	89** Ac 1.10	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

* Lanthanides	58 Ce 1.12	59 Pr 1.13	60 Nd 1.14	61 Pm	62 Sm 1.17	63 Eu	64 Gd 1.20	65 Tb 1.21	66 Dy 1.22	67 Ho 1.23	68 Er 1.24	69 Tm 1.25	70 Yb	71 Lu 1.27
** Actinides	90 Th 1.30	91 Pa 1.50	92 U 1.38	93 Np 1.36	94 Pu 1.28	95 Am 1.30	96 Cm 1.28	97 Bk 1.30	98 Cf 1.30	99 Es 1.30	100 Fm 1.30	101 Md 1.30	102 No 1.30	103 Lr 1.30

atomic number
Symbol
electronegativity

Pauling scale. Noble gases are left blank. Blank cells elsewhere have no measured value.

Polyatomic Ions

CHEMISTRY REFERENCE

★ **Memorize these 15.** The rest are here to look up. Each group lists the ions to memorize first.

1+ ions

★	**ammonium**	**NH_4^+**
★	**hydronium**	**H_3O^+**

2+ ions

	mercury(I)	Hg_2^{2+}

1– ions

★	**acetate**	**$C_2H_3O_2^-$**
★	**hydrogen carbonate (bicarbonate)**	**HCO_3^-**
★	**chlorate**	**ClO_3^-**
★	**cyanide**	**CN^-**
★	**hydroxide**	**OH^-**
★	**nitrate**	**NO_3^-**
★	**nitrite**	**NO_2^-**
★	**permanganate**	**MnO_4^-**
	perchlorate	ClO_4^-
	chlorite	ClO_2^-
	hypochlorite	ClO^-
	bromate	BrO_3^-
	iodate	IO_3^-
	periodate	IO_4^-
	hydrogen sulfate (bisulfate)	HSO_4^-
	hydrogen sulfite (bisulfite)	HSO_3^-
	hydrogen sulfide (bisulfide)	HS^-
	dihydrogen phosphate	$H_2PO_4^-$
	thiocyanate	SCN^-
	cyanate	OCN^-

2– ions

★	**carbonate**	**CO_3^{2-}**
★	**chromate**	**CrO_4^{2-}**
★	**sulfate**	**SO_4^{2-}**
★	**sulfite**	**SO_3^{2-}**
	dichromate	$Cr_2O_7^{2-}$
	hydrogen phosphate	HPO_4^{2-}
	oxalate	$C_2O_4^{2-}$
	peroxide	O_2^{2-}
	thiosulfate	$S_2O_3^{2-}$
	silicate	SiO_3^{2-}

3– ions

★	**phosphate**	**PO_4^{3-}**
	phosphite	PO_3^{3-}
	arsenate	AsO_4^{3-}
	borate	BO_3^{3-}

Naming Toolkit

CHEMISTRY REFERENCE

Roman Numerals

The charge on a metal ion in a name: iron(III) is Fe^{3+}.

Numeral	Value
I	1
II	2
III	3
IV	4
V	5
VI	6
VII	7
VIII	8

Covalent Prefixes

The number of each atom in a molecular compound.

Prefix	Number
mono-	1
di-	2
tri-	3
tetra-	4
penta-	5
hexa-	6
hepta-	7
octa-	8
nona-	9
deca-	10

Diatomic Elements

These seven exist as two-atom molecules when the element is on its own.

H_2 N_2 O_2 F_2 Cl_2 Br_2 I_2

Metals with Multiple Charges

CHEMISTRY REFERENCE

These metals form more than one ion. The Roman numeral in the name gives the charge.

Metal	Ion		Ion	
Copper	**Cu^{+}**	copper(I)	**Cu^{2+}**	copper(II)
Iron	**Fe^{2+}**	iron(II)	**Fe^{3+}**	iron(III)
Lead	**Pb^{2+}**	lead(II)	**Pb^{4+}**	lead(IV)
Tin	**Sn^{2+}**	tin(II)	**Sn^{4+}**	tin(IV)
Mercury	**Hg_2^{2+}**	mercury(I)	**Hg^{2+}**	mercury(II)
Cobalt	**Co^{2+}**	cobalt(II)	**Co^{3+}**	cobalt(III)
Nickel	**Ni^{2+}**	nickel(II)	**Ni^{3+}**	nickel(III)
Chromium	**Cr^{2+}**	chromium(II)	**Cr^{3+}**	chromium(III)
Manganese	**Mn^{2+}**	manganese(II)	**Mn^{3+}**	manganese(III)
Gold	**Au^{+}**	gold(I)	**Au^{3+}**	gold(III)

Transition metals with one charge

Silver	**Ag^{+}**	always this charge — no Roman numeral
Zinc	**Zn^{2+}**	always this charge — no Roman numeral
Cadmium	**Cd^{2+}**	always this charge — no Roman numeral

Solubility

CHEMISTRY REFERENCE

cations ↓ \ anions →	NO_3^- nitrate	$C_2H_3O_2^-$ acetate	ClO_3^- chlorate	Cl^- chloride	Br^- bromide	I^- iodide	SO_4^{2-} sulfate	OH^- hydroxide	CO_3^{2-} carbonate	PO_4^{3-} phosphate	S^{2-} sulfide
NH_4^+ ammonium	S	S	S	S	S	S	S	S	S	S	S
Na^+ sodium	S	S	S	S	S	S	S	S	S	S	S
K^+ potassium	S	S	S	S	S	S	S	S	S	S	S
Mg^{2+} magnesium	S	S	S	S	S	S	S	I	I	I	X
Ca^{2+} calcium	S	S	S	S	S	S	*SS*	*SS*	I	I	X
Sr^{2+} strontium	S	S	S	S	S	S	I	*SS*	I	I	X
Ba^{2+} barium	S	S	S	S	S	S	I	S	I	I	X
Al^{3+} aluminum	S	S	S	S	S	S	S	I	X	I	X
Mn^{2+} manganese(II)	S	S	S	S	S	S	S	I	I	I	I
Fe^{2+} iron(II)	S	S	S	S	S	S	S	I	I	I	I
Fe^{3+} iron(III)	S	S	S	S	S	X	S	I	X	I	X
Co^{2+} cobalt(II)	S	S	S	S	S	S	S	I	I	I	I
Ni^{2+} nickel(II)	S	S	S	S	S	S	S	I	I	I	I
Cu^{2+} copper(II)	S	S	S	S	S	X	S	I	I	I	I
Zn^{2+} zinc	S	S	S	S	S	S	S	I	I	I	I
Ag^+ silver	S	*SS*	S	I	I	I	*SS*	I	I	I	I
Hg_2^{2+} mercury(I)	S	*SS*	S	I	I	I	I	I	I	I	I
Pb^{2+} lead(II)	S	S	S	*SS*	*SS*	I	I	I	I	I	I

S soluble (more than 1 g per 100 mL)

SS slightly soluble (0.1–1 g)

I insoluble (less than 0.1 g) — a precipitate

X reacts with water or does not form

Water at about 25 °C.

Rules

1. Compounds of Group 1 ions and NH_4^+ are soluble. So are all nitrates and chlorates, and nearly all acetates.
2. Chlorides, bromides and iodides are soluble, except with Ag^+, Hg_2^{2+} and Pb^{2+}.
3. Sulfates are soluble, except with Sr^{2+}, Ba^{2+}, Pb^{2+} and Hg_2^{2+}. Ca^{2+} and Ag^+ sulfates are slightly soluble.
4. Hydroxides are insoluble, except with Group 1, NH_4^+ and Ba^{2+}. Ca^{2+} and Sr^{2+} hydroxides are slightly soluble.
5. Carbonates, phosphates and sulfides are insoluble, except with Group 1 and NH_4^+. Group 2 sulfides react with water.
6. **X** cells break these rules on purpose: those ions react with each other or with water instead of forming the compound. Check the grid.

Activity Series

CHEMISTRY REFERENCE

In a single-replacement reaction, an element replaces any element **below** it in its own list. It does not replace an element above it.

Metals

most active

- Li
- K
- Ba
- Sr
- Ca
- Na

Reacts with cold water, steam and acids

- Mg
- Al
- Mn
- Zn
- Cr
- Fe
- Cd

Reacts with steam and acids

- Co
- Ni
- Sn
- Pb

Reacts with acids

- H_2

- Cu
- Hg
- Ag
- Pt
- Au

Does not replace hydrogen from acids

least active

Halogens

most active

- F_2
- Cl_2
- Br_2
- I_2

least active

Strong Acids & Bases

CHEMISTRY REFERENCE

Strong acids and bases ionize completely in water. Treat any acid or base not on this sheet as weak.

Strong Acids

HCl	hydrochloric acid
HBr	hydrobromic acid
HI	hydroiodic acid
HNO_3	nitric acid
H_2SO_4	sulfuric acid
$HClO_4$	perchloric acid

Strong Bases

LiOH	lithium hydroxide
NaOH	sodium hydroxide
KOH	potassium hydroxide
RbOH	rubidium hydroxide
CsOH	cesium hydroxide
$Ca(OH)_2$	calcium hydroxide
$Sr(OH)_2$	strontium hydroxide
$Ba(OH)_2$	barium hydroxide

Group 1 hydroxides, and the hydroxides of Ca, Sr and Ba.

Specific Heats

CHEMISTRY REFERENCE

Specific heat (c) is the energy needed to raise 1 g of a substance by 1 °C. Use it in q = mcΔT (Equations sheet).

Substance	c, J/(g·°C)
Water (liquid)	**4.184**
Ice	**2.06**
Steam	**2.01**
Ethanol	**2.44**
Aluminum	**0.897**
Glass	**0.84**
Granite	**0.79**
Iron	**0.449**
Nickel	**0.444**
Zinc	**0.388**
Copper	**0.385**
Brass	**0.380**
Silver	**0.235**
Tin	**0.228**
Lead	**0.129**
Gold	**0.129**